



# RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Computer Science and Engineering

Academic Year 2025 – 2026 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. CSE

Course Code & Title: CS3551 Distributed Computing

Name of the Faculty member (s): Mrs.P.Devisri

Innovative Practice Description

Unit / Topic: Unit I / Global State of Distributed System

Course Outcome: CO 1

Unit Learning Outcome: 1a

Activity Chosen: Collaborative Learning Activity

## Justification:

- The activity aims to help students understand the concept of **Global State** in Distributed Systems through teamwork and group discussions.

- This practice was conducted to enable students to learn collaboratively, share ideas, and explore how different processes in a distributed system work together without a common clock or single controller.

- The session aimed to bridge theoretical concepts with practical problem-solving, enhancing analytical thinking, collaboration, and understanding of system consistency in distributed computing.

- Time Allotted for the Activity: **30 minutes**

## Details of Implementation:

### 1. Formation of Teams:

Students from **III Year CSE – B** were divided into **four small groups** (4–5 members each).

- **Team 1:** Kathirvel N S, Dharsan sivakumar, Hari Krishnan C, Mohan Ganesh
- **Team 2:** Nivedaa K, Nafeessa, Pushpakalai S, Subhashri M
- **Team 3:** Saravana Sanjay M, Saravanan A, Dhinesh Araving G, BharathRaj J
- **Team 4:** Rasiga S, Anu Keerthana M, Dhivya Bharathi G, Mariammal G

### 2. Concept Introduction:

The teacher first explained what a **Global State** is and why it is important in a distributed system.

Key points like **local state**, **channel state**, and the **Chandy-Lamport Snapshot Algorithm** were introduced with a simple example.

### 3. Group Activity:

Each group was given a small distributed system example.

They had to:

- Write down the local state of each process.
- Find the messages that were sent but not yet received.
- Check if their combined snapshot was a consistent global state.
- Draw a simple diagram to show process and message flow.

### 4. Group Discussion:

Each team worked together to find the correct answers.

- Team 2 first made a mistake in message order, but after a discussion with Team 3, they corrected it and understood the concept better.

### 5. Presentation and Feedback:

Every team presented their diagrams and results.

- Team 4 explained clearly how they captured messages in transit to maintain consistency.

The teacher gave feedback, corrected small mistakes, and praised the teamwork and effort of all students.

Images / Screenshot of the practice:



CO – PO / PSO mapping:

| CO  | PO1 | PO2 | PO3 | PO4 | PO5 | PO9 | PO10 | PSO1 |
|-----|-----|-----|-----|-----|-----|-----|------|------|
| CO1 | 2   | 2   | 2   | 1   | 1   | 1   | 1    | 1    |

(1 – Low 2 – Moderate 3 – High)

**PO / PSO mapped:**

| Innovative practice                  | PO1                                                                                                                                        | PO2                                                                                                                                         | PO3                                                                                                                             | PO4                                                                                                                                                   | PO5                                                                                                                                         | PO9                                                                                                                             | PO10                                                                                                                                | PSO1                                                                                                                                                     |
|--------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                      | 2                                                                                                                                          | 2                                                                                                                                           | 2                                                                                                                               | 1                                                                                                                                                     | 1                                                                                                                                           | 1                                                                                                                               | 1                                                                                                                                   | 1                                                                                                                                                        |
| <b>Justification for correlation</b> | Students gained a clear understanding of the concept of Global State and its importance in maintaining consistency in distributed systems. | Students analyzed distributed process interactions and message passing to identify consistent states through collaborative problem-solving. | Students designed and developed state diagrams and snapshots to represent consistent global states in distributed environments. | Students applied their theoretical knowledge to a realistic distributed system scenario, identifying challenges in synchronization and message order. | Students demonstrated the use of logical thinking and algorithmic approach in computing consistent global states and verifying correctness. | Students effectively communicated within teams, shared ideas and clarified doubts, improving teamwork and collaboration skills. | Students presented their findings through group presentations and peer feedback, enhancing their technical communication abilities. | Students applied their understanding of distributed system concepts and snapshot algorithms to real-world computing problems using analytical reasoning. |

**Reflective Critique:**

*Feedback of practice from students and other stakeholders:*

- Students shared that this activity provided a clear and practical understanding of the **Global State** concept in Distributed Systems.
- They appreciated the group discussions and teamwork, which helped them visualize how different processes and messages are recorded together in a distributed system.
- The activity encouraged active participation, problem-solving, and communication among peers, making the learning experience more engaging and interactive.

*Benefit of the Practice:*

- Students developed a clearer understanding of the **Global State** concept.
- They improved teamwork, communication, and problem-solving skills.

- All students participated actively and showed greater interest in the topic.
- Students became more confident in explaining distributed system concepts.

*Challenges Faced in Implementation:*

- Some teams initially made mistakes in message ordering or process analysis, requiring additional discussion to correct.
- Ensuring equal engagement and participation from all students in each group was sometimes challenging.